Dipartimento di Ingegneria e Scienza dell'Informazione

– KnowDive Group –

Knowledge Graph course 2025 - Project Report

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1	9 Nov 2025	Mattia Santaniello	Document created
2	9 nov 2025	Mattia Santaniello	Personas, Scenarios and Competency questions
3	11 Nov 2025	Mattia Santaniello	ER model 1st version
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5	12 Gen 2026	Mattia Maci and Mattia Santaniello	Concept identification and ER remodeled, 2nd version
6	14 Gen 2026	Mattia Maci and Mattia Santaniello	Remodelled, 3rd version
7	15 Gen 2026	Mattia Maci	Information gathering
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1 Introduction

Reusability is one of the main principles in the Knowledge Graph (KG) development process defined by iTelos. The KG project documentation plays an important to enhance the reusabiltiy of the resources handled and produced during the process. A clear description of the resources, the process (and sub processes) developed and evaluation at each step of the process provides a clear understanding of the project, thus serving such an information to external readers for the future exploitations of the project's outcomes.

The current document aims to provide a detailed report of the project developed following the iTelos methodology. The report is structured, to describe:

- Section 2: Definition of the project's purpose and related information gathering.
- Sections 3, 4, 5, 6: The description of the iTelos process phases and their activities, divided by knowledge and data layer activities, as well as the evaluation of the resources produced in terms of fit for the chosen purpose.
- Section 7: The description of the metadata produced for all (and all kind of) the resources handled and generated by the iTelos process, while executing the project.
- Section 8: Conclusion and open issues summary.

2 Project Design

This project is about building a Knowledge Graph that brings together data from different weather stations to better understand the climate of the Trentino area during the decade 2015-2025. It helps organize and connect meteorological data in a smart way so users can easily explore trends and patterns. Using this graph, you can ask meaningful questions about temperature, rainfall, and changes over time, like: We can see how our domain of interest includes meteorological observations, spatial information about weather stations, and temporal aspects related to climate evolution over time.

3 Purpose Definition

In the following section we are taking the informal description of the purpose in the project design and use *iTelos* methodology to extract a formal description of the purpose.

3.1 Domain

The area of knowledge and the field of study interested are climatology and environmental data analysis, with a specific focus on regional climate change. More precisely, the project lies at the intersection of meteorology, geospatial analysis, and semantic data modeling through Knowledge Graphs(KG).

3.2 Context

more specifically we are interested in the context of the Trentino area, during the decade 2015-2025, analyzing local climate variations, weather patterns, and their impact on the environment and communities.

3.3 Scenarios

now we will continue illustrating the possible scenarios:

- **S1: Mario Rossi** is performing a research on the effects of climate change, which could be useful for some supervisor agencies specialized on the theme (e.g ESA). He needs to retrieve some data about the weather of the territory, analyzing how the temperature has evolved over the years and how much it has drifted from the natural trend.
- **S2: Laura Rosa** is planning a long journey on the mountains with her friends, and she needs to know the meteorological data of the locations they are going to visit, especially the rainfall events and the sunny days of the previous years in order to make decision on the logistics
- **S3: Luigi Verdi** is worried about how the climate events of the last years could affect his farming activities and damage his products.Agriculture is one of the sectors which has been mostly affected negatively by the climate change, so he wants to be prepared

3.4 Personas

Now that the scenarios have been defined it is time to see the profiles of the users mentioned above:

- **P1: Mario Rossi** is a 26 years old researcher at the Trento university, specialized in ecology and enviromental research
- **P2: Laura Rosa** is a 30 years old woman who is also planning a trip on the mountains of the Trentino region, however in her case the main activity is a trekking exursion with her friends
- **P3: Luigi Verdi** is a 56 years old farmer, with a small field nearby Trento. His main products are potatoes and grape, the latter one especially is cultivated in summer and gathered during grape harvest period

3.5 Competency Question

Each Persona has to deal with different scenarios, which could be encapsulated in the definition of Competency Questions(CQ). Some of them could be formalized in the following table

Person	Question
Mario Rossi	**Q1:** What is the trend of the temperature values in Trentino during the decade 2015-2025?
Mario Rossi	**Q2:** How the metheorogical facilities are distributed in the territory?
Mario Rossi	**Q3:** Which areas in Trentino have seen the highest increase in temperature over time?
Laura Rosa	**Q4:** How was the temperature on the Trentino mountains during the winter of the last three years?
Laura Rosa	**Q5:** What are the locations where rainfalls most frequently happen?
Laura Rosa	**Q6:** What are the periods of the year where there are more sun radiations?
Luigi Verdi	**Q7:** Did the number of extreme climate events increased in the area around my field?
Luigi Verdi	**Q8:** What is the trend of humidity in the summer of the last three years?
Luigi Verdi	**Q9:** Did the number of rainfalls decreased over the last five years?
Mario Rossi	**Q10:** Did the number atmospheric pressure changed over the last decade?

Table 1: Competency Questions

3.6 Purpose Formalization

now from the scenarios, the personas, and the competency questions we extract a list of useful concepts, During the concept identification phase, we focused on concepts that are directly required to answer the defined competency questions and that can be grounded in the available data sources.

Full file in PurposeFormalization.xlsx

Starting from the concept of **Meteorological Facility** and **Meteorological Measurement**, which are the most prominent entities in the dataset, we modeled all the other concepts.

About the **Meteorological Facility** we had data about the coordinates, so we decided to model the concept of **coordinates** and consequently the concept of **Bacino**, being the station located inside a ”Bacino” in our dataset, and the concept of **Trentino**.

Similarly for the **Meteorological Measurement** we modeled the concept of **Month** and **TimeInterval**, representing the moment, with different granularity, in which the measurement

Scenarios	Personas	Competency Questions	ID	Entities	Properties	Focus
S1,S2,S3	P1,P2,P3	Q1,Q2,Q3,Q4,Q5,Q6,Q7,Q8,Q9, Q10	Q190107	Meteorological facility	MeteorologicalFacilityID, Stazione, Tavoletta n., Note	Common
S1,S2,S3	P1,P2,P3	Q1,Q3,Q4,Q5,Q6,Q7,Q8,Q9, Q10	UNITN2026KGMMC01	Meteorological measurement	MeteorologicalMeasurementID, Val, unit	Contextual
S1,S2	P1,P2	Q1,Q3,Q4	UNITN2026KGMMC02	TempMeasurement	TempMeasurementID, statisticType	Contextual
S2,S3	P2,P3	Q5,Q7,Q9	UNITN2026KGMMC03	RainMeasurement	RainMeasurementID	Contextual
S3	P3	Q8	UNITN2026KGMMC04	HumidityMeasurement	HumidityMeasurementID	Contextual
S3	P3	Q7	UNITN2026KGMMC05	WindMeasurement	WindMeasurementID, Type	Contextual
S1	P1	Q10	UNITN2026KGMMC06	PressureMeasurement	PressureMeasurementID	Contextual
S2	P2	Q6	UNITN2026KGMMC07	SolarRadiationMeasurement	SolarRadiationMeasurementID	Contextual
S1,S2,S3	P1,P2,P3	Q1,Q2,Q3,Q4,Q5,Q6,Q7,Q8,Q9, Q10	Q5151	Month	MonthValue	Common
S1,S2,S3	P1,P2,P3	Q1,Q2,Q3,Q4,Q5,Q6,Q7,Q8,Q9, Q10	Q186081	TimeInterval	Start, End	Common
S1,S2,S3	P1,P2,P3	Q1,Q2,Q3,Q4,Q5,Q6,Q7,Q8,Q9, Q10	Q22664	Coordinate	CoordinateID, Coordinate Est/ Nord, Latitudine, Longitudine	Common
S1,S2,S3	P1,P2,P3	Q1,Q2,Q3,Q4,Q5,Q6,Q7,Q8,Q9, Q10	Q39816	Bacino	NomeBacino	Common
S1,S2,S3	P1,P2,P3	Q1,Q2,Q3,Q4,Q5,Q6,Q7,Q8,Q9, Q10	Q16289	Trentino	TrentinoID	Common
S1,S2,S3	P1,P2,P3	Q1,Q3,Q4,Q5,Q6,Q7,Q8,Q9, Q10	Q167676	Sensor	SensorID	Common
S3	P3	Q7	Q1277161	Extreme events	ExtremeEventsID	Common
S1,S2,S3	P1,P2,P3	Q1,Q3,Q5,Q6,Q7,Q8,Q9, Q10	UNITN2026KGMMC08	Trend	TrendID	Core
S1,S2,S3	P1,P2,P3	Q1,Q3,Q5,Q6,Q8,Q9, Q10	UNITN2026KGMMC09	Climate change indicator	ClimateChangeIndicatorID	Core
S1,S2,S3	P1,P2,P3	Q1,Q3,Q5,Q6,Q8,Q9, Q10	Q125928	Climate change	ClimateChangeID	Common

Figure 1: Purpose formalization overview

took place.

having different types of **Meteorological Measurement**, we modeled the hierarchical concepts of **TempMeasurement**, **RainMeasurement**, **HumidityMeasurement**, **WindMeasurement**, **PressureMeasurement**, **SolarRadiationMeasurement**.

To ontologically split the concept of **Meteorological Facility** and **Meteorological Measurement** we modeled the concept of **Sensor**, so the **Meteorological Facility** contains **Sensors**, which performs **Meteorological Measurement**. We modeled the concept of **Trend**. **Trend** is a **Climate change indicator**, which is pert of **Climate change**.

Some concepts refer to concepts already modeled in wikidata, therefore for this concept we linked the ID to the wikidata one, for their nature the concept with contextual focus are not very reusable so cannot be linked to preexisting concepts

3.7 ER Diagram

At this point the next step of the formalization procedure is to produce a first bone structure which allows us to define the involved entities more clearly. Lets first define their schemas:

full diagram in ER.drawio

In Figure 1, the core entities of the Knowledge Graph represent the fundamental components required to describe and analyze meteorological data at a regional scale.

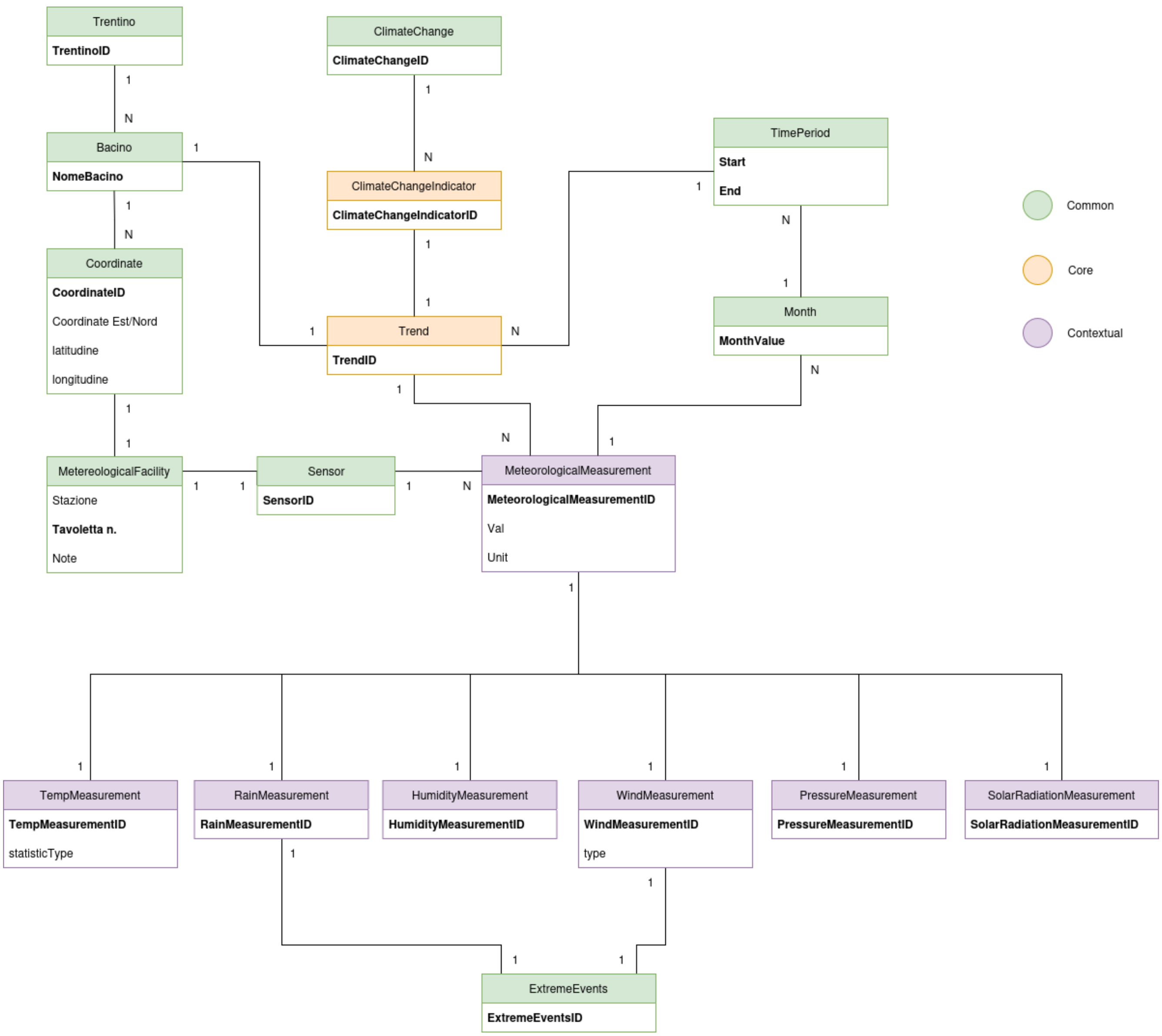


Figure 2: ER diagram

The entity **Trentino** represents the Trentino region of Italy, while **Bacino** models the main valley within this region. **Coordinate** represents specific points in space corresponding to the location of meteorological facilities.

The entity **Meteorological facility** represents a weather station, which hosts one or more **Sensors**, each capable of observing specific meteorological phenomena. **Meteorological measurement** represents a generic observation, with temporal information captured by **Month** and **TimeInterval**. Specific types of measurements, such as temperature (average, minimum, or maximum, recorded via *statisticType*), rainfall, humidity, wind (speed and direction), pressure, and solar radiation, are modeled as specialized entities linked to Meteorological measurement, **ExtremeEvent** represents unusually high or low values of rainfall or wind.

Trend: Each Trend instance represents a concrete climate indicator computed for a specific geographic area and a specific time interval, computed from Meteorological measurements and representing long-term variations. Each **Trend** instance is computed from more **Meteorological measurement** and is a **ClimateChangeIndicator**, which in turn is part of **ClimateChange**, ClimateChange is modeled as a high-level abstract concept, not directly associated with specific observations, but represented through derived indicators such as Trends.

3.8 Information Gathering

here we provide an overview of the data used in input for this project, including resources using different languages. We want to gather relevant information coherent with our purpose, in order to construct a meaningful knowledge graph. The knowledge layer of the proposed Knowledge Graph is grounded on external semantic resources. In particular, Wikidata is employed as the primary external knowledge source to link selected domain entities, such as Meteorological facility (or weather station for sinonimicity), the region of Trentino and so on, to well-established real-world concepts. Wikidata offers a collaboratively maintained, structured knowledge base that ensures semantic consistency and interoperability with existing linked data resources.
These resources are not directly reused as ontologies but are referenced through semantic links to provide an external grounding for key concepts. The adopted formal knowledge resources can be classified as **common** resources, as Wikidata provides general-purpose concepts widely used across multiple domains, No core or contextual domain-specific ontologies were directly reused. Additional domain ontologies were considered during the design phase but were not directly adopted in order to preserve model simplicity and avoid unnecessary complexity. The data layer of the Knowledge Graph is built on meteorological measurements and weather stations collected from the open-access repository meteotrentino, which provides detailed information about weatherstations in the Trentino region, including geographic coordinates, macro-areas and station ID and name. Data fields are composed by all of different measurement categories such as temperature, rainfall, humidity, wind, pressure, and solar radiation, corresponding to the main meteorological phenomena relevant to climate analysis, and relative timestamp.

3.8.1 MetereologicalFacility

MetereologicalFacility, or weather station, represents the building located in a specific point of the Trentino region, this resource has the following attributes.

- **Stazione**: (String) the name of the station
- **Tavoletta n.**: (Integer) the unique identifier of the station
- **Note**: (String) some additional information about hte station

the attributes are in italian, because the data was in italian in the website we got them.
The focus is **common** because it is reusable and we linked it with a Wikidata concept

3.8.2 Coordinate

Is the poin inside of the Trentino region in which a station is located.
Key attributes are

- **CoordinateID**: (Integer) the unique identifier of the space point
- **Coordinate Est/Nord** (String)represents the distance from the north pole and from the main meridian, separated by a ”/”
- **latitudine** (String) represents the latitude of the point
- **longitudine** (String) represent the longitude of the point

the attributes and the name are in italian, because the data was in italian in the website we got them.
The focus is **common** because it is reusable and we linked it with a Wikidata concept

3.8.3 Bacino

Is a valley inside the Trentino region.
Key attributes

- **NomeBacino**: (String) the name of the valley, also the unique identifier

the attributes and the name are in italian, because the data was in italian in the website we got them
The focus is **common** because it is reusable and we linked it with a Wikidata concept

3.8.4 Trentino

Is the region in Italy we are focusing out project on.
Key attributes

- **TrentinoID**: (Integer) unique identifier

The focus is **common** because it is reusable and we linked it with a Wikidata concept

3.8.5 Sensor

Is the hardware device used to perform the actual weather measurement, is sited inside a weather station.
Key attributes

- **SensorID**: (integer) unique identifier

The focus is **common** because it is reusable and we linked it with a Wikidata concept

3.8.6 MetereologicalMeasurement

The actual measurement of a weather observable.
Key attributes

- **MetereologicalMeasurementID**: (Integer) unique identifier
- **Value**: (Integer) the value of the measurement
- **Unit**: (String) the unit of measurement

The focus is **contextual** because is to specific to be reused, and we couldn't find any fitting preexisting concept

3.8.7 TempMeasurement

A specific kind of MetereologicalMeasurement, representing the measurement of a temperature value.
Key attributes

- **TempMeasurementID**: (Integer) unique identifier
- **statisticType**: (String) Determine if the value is a average, a minimum or a maximum

The focus is **contextual** because is to specific to be reused, and we couldn't find any fitting preexisting concept

3.8.8 RainMeasurement

A specific kind of MetereologicalMeasurement, representing the measurement of a rain value.
Key attributes

- **RainMeasurementID**: (Integer) unique identifier

The focus is **contextual** because is to specific to be reused, and we couldn't find any fitting preexisting concept

3.8.9 HumidityMeasurement

A specific kind of MetereologicalMeasurement, representing the measurement of a humidity value.
Key attributes

- **HumidityMeasurementID**: (Integer) unique identifier

The focus is **contextual** because is to specific to be reused, and we couldn't find any fitting preexisting concept

3.8.10 WindMeasurement

A specific kind of MetereologicalMeasurement, representing the measurement of a wind value.
Key attributes

- **WindMeasurementID**: (Integer) unique identifier
- **Type**: (String) Determine if the value refers to the speed or the direction

The focus is **contextual** because is to specific to be reused, and we couldn't find any fitting preexisting concept

3.8.11 PressureMeasurement

A specific kind of MetereologicalMeasurement, representing the measurement of a pressure value.
Key attributes

- **PressureMeasurementID**: (Integer) unique identifier

The focus is **contextual** because is to specific to be reused, and we couldn't find any fitting preexisting concept

3.8.12 SolarRadiationMeasurement

A specific kind of MetereologicalMeasurement, representing the measurement of a solar radiation value.
Key attributes

- **SolarRadiationMeasurementID**: (Integer) unique identifier

The focus is **contextual** because is to specific to be reused, and we couldn't find any fitting preexisting concept

3.8.13 ExtremeEvent

A specific kind of RainMeasurement or WindMeasurement with a very high value.
Key attributes

- **ExtremeEventID**: (Integer) unique identifier

The focus is **Common** because even if is a specific kind of a contextual Etype, this is modeled in Wikidata

3.8.14 Month

Irregular unit of time dividing a calendar year.
Key attributes

- **MonthValue**: (String) The name of the month, also the unique identifier

The focus is **common** because it is reusable and we linked it with a Wikidata concept

3.8.15 TimePeriod

temporal extent having a beginning, an end and a duration.
Key attributes

- **Start**: (String) The moment from which the time interval starts, part of unique identifier
- **End**: (String) The moment in which the time interval ends, part of unique identifier

The focus is **common** because it is reusable and we linked it with a Wikidata concept

3.8.16 Trend

cluster of statistical values showing how a variable is changing in time.
Key attributes

- **TrendID**: (Integer) unique identifier

The focus is **core** because it is reusable bat to specific and we didn't find a 1 to 1 correspondence in wikidata

3.8.17 ClimateChangeIndicator

A trend that shows how the climate is changing.
Key attributes

- **ClimateChangeIndicatorID**: (Integer) unique identifier

The focus is **core** because it is reusable bat to specific and we didn't find a 1 to 1 correspondence in wikidata

3.8.18 ClimateChange

human-caused changes to climate on Earth.
Key attributes

- **ClimateChangeID**: (Integer) unique identifier

The focus is **common** because it is reusable and we linked it with a Wikidata concept

3.9 Evaluation - Purpose Definition

We faced some issue during the purpose definition phase, mainly we had to change the modeling, because in the beginning it was under modelled, and after this we made it more complex, reaching a model that we judged as over modelled, finding in the end a good middle ground. But still there are some peculiarity that we choose to keep for eventual future modification, for instance the Etype PressureMeasurement was not answering any competency question, so we decided to add a specific competency question to make it more integrated in the model, we added the competency question identified by ID Q10. With the data finally retrieved we can see that not all competency questions can be answered: some of them in fact, like Q3 and Q5, can be satisfied only if we considered the geographical area where the measurements have been made, otherwise it would be impossible to answer them since the locations data were unavailable, also we haven't been able to retrieve data in order to classify weather conditions as extreme or not; for these reason we decided to remove the competency questions Q4 and Q7 since there are no data that allow satisfy them. Problems arised also during the retrievement of the data from different stations: it has been seen that it is not possible to automatize the process through scripting because the download is made by a dynamically generated link that changes every time a data request spreadsheet is sent to the website; for this reason we will consider only an arbitrary subset of meteorological facilities spread across Trentino, while the other ones will be considered non existsent.

4 Language Definition

This section describes the language definition phase. This phase is fundamental as it ensures clarity and consistency by addressing ambiguity and diversity in natural and domain-specific languages. In particular, it explicitly takes into account the heterogeneity of the adopted resources, since some of the consulted datasets and references are available in Italian, while others are provided in English.

By defining a shared domain language, this phase enables accurate data annotation, minimizes misunderstandings arising from linguistic variations, and supports seamless integration across systems and sources. This step is therefore crucial for effective communication and interoperability in complex, multilingual and multi-domain projects.

4.1 concept identification

going back to concept identification done in the purpose definition, a domain language ahs been developed. This has been done by associating a unique identifier to each concept, and by giving a short definition of each concept for clarity. The domain language is avaliabe in LenguageFormalization.xlsx.

In this activity we focused on aligning the developed concept to already modeled concepts present on Wikidata, to ground them on external semantic resources.

Some concept did not match with preexisting ones, so we created custom definitions tailored to our purpose.

The same been done with the data property and the object property.

In this phase we have taken charge of the problem of the multilingual resources, we decided to have all the information in one language, english.

Is also at this layer that we took care of the polysemy and sinonimicity.

5 Knowledge Definition

The knowledge definition phase of the *iTelos* methodology consists on the construction of the knowledge graph by formalizing and aligning the information gathered so far and create a unified knowledge representation via a teleteleology. Starting from the output of the previous layer we want to build an extended entity relation model (EER), which points to reflect a more clear vision of the classic ER previously defined by implementing the concept of generalization: we define specific entities which represents particular istances of a more generalized superclass; this process is done by introducing the **'is-a' relation**, which explicit the hierarchy of the entity types. The ER model must be modified by taking advantage from the developed domain language, adding possible class inheritances, which results in the figure 3, that can also be found here

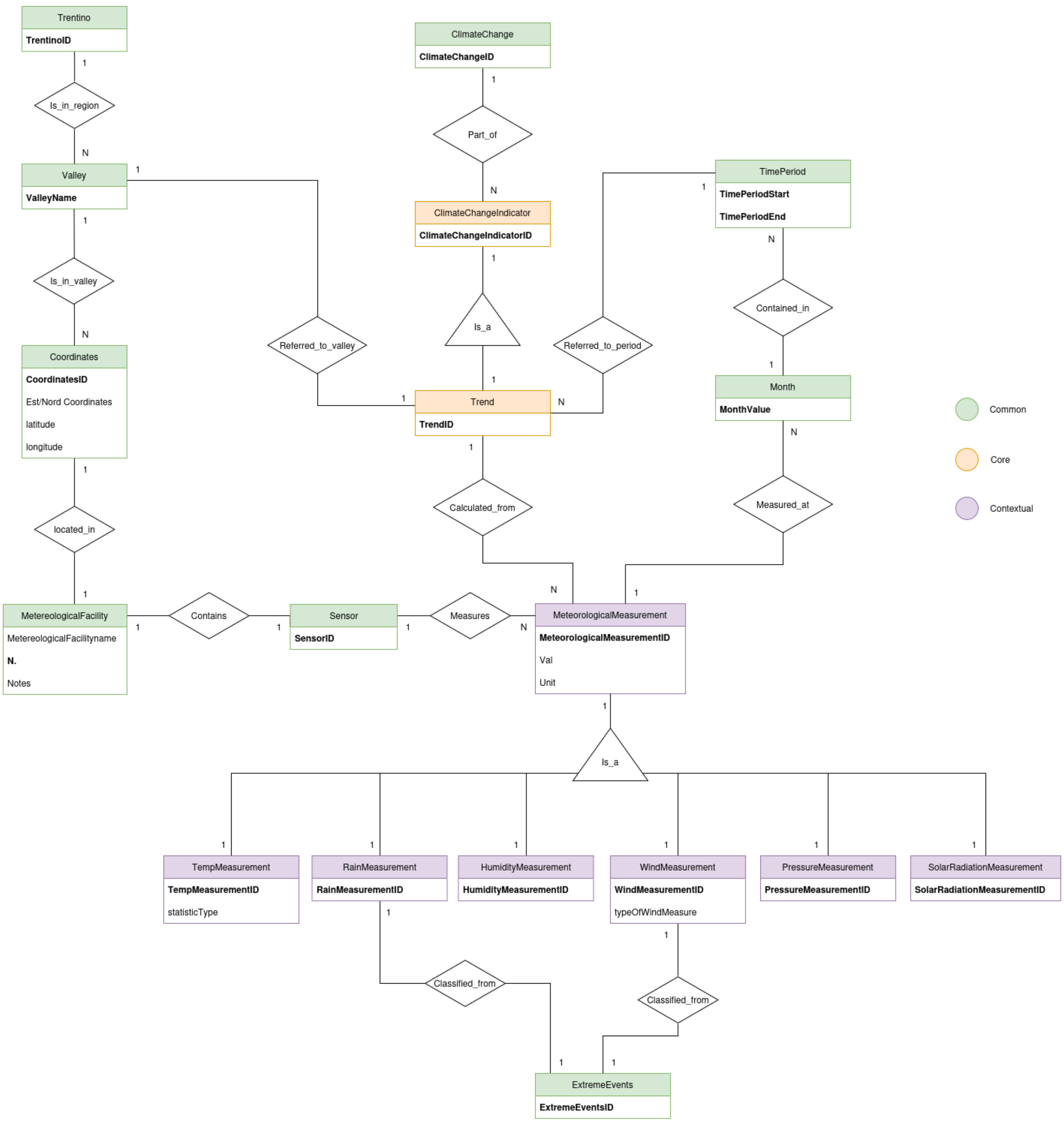


Figure 3: EER diagram

Although the EER diagram is based on the previous Language Phase is important to notice that it still lacks a relevant piece that without its presence does not allow do encapsulate any kind of information: conceptualization. by using Protègè, a specific tool that through its UI allowed us to define entity types, object properties and data properties, to build a solid teleology, and to link our work to a well established set of concepts. The result of this work can be found in UNITN2026KGMM_Knowledge_definition.ofn. in Protègè we started by creating all the entity types, ang assigning to each a specific concept ID, as seen in the language definition phase some concept was tailored made for this project, like TempMeasurement, while some other are referred to wikidata concepts, Wikidata was adopted as reference knowledge resource for concept alignment, due to its rich coverage of geographical entities and environmental concepts relevant to the project domain, for these concepts we populated the exactMatch annotation with the link to the wikidata page, we modeled the hierarchy as shown in the EER model, we also modeled Climate change indicator as contained in the climate change concept, and extreme event as a part of the union of WindMeasurement and RainMeasurement.

We did the same with data properties populating also the domain and range records, the domain is the entity type the data property is referred to, and the range is eather Integer, Float or String.

Lastly for the object properties again we modelled all the ID, and the domain and range records, where the domain is the subject of the relationship and the range is the object. In figure 4 we can see a piece of the representation, showing the Etypes hierarchy

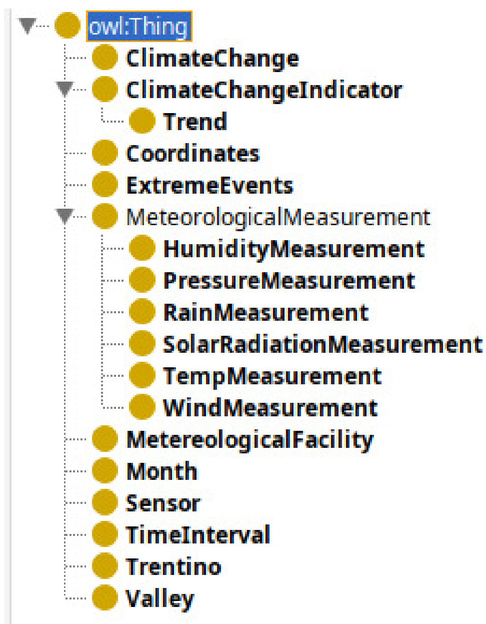


Figure 4: Protegeetypes

We can notice how the structure of the teleontology is pretty much the same of the original ER schema, however thanks to the definition of the EER diagram and the introduction of is-a relationships it results to have a much more organized structure thanks to the implemented hierarchy.

6 Entity Definition

With the results obtained from previous chapters we have created a teleontology that represent an abstract conceptualization of the real world we want to interpet; however it is necessary now to take care of the **meaning hetereogenity**: we need to find a way to distinguish two entities based not only on the abstract conecpt represented by the teleontology but also we need to consider the possible diversity of the datasets involved. This is the final spet in *iTelos* metodology, merging the knowledge and data layers into a unified Knowledge Graph, the input are the data resources and the teleontology developed in previous phase. To do so, we have to teke care of the heterogeneity in the data layer, in this layer heterogeneity is shown in 3 different ways

- **Entity Matching**: resolves discrepancies in the representation of real-world entities across different datasets, considering both schema and data-level heterogeneity.
- **Entity Identification**: ensures each entity is uniquely identified, using either existing identifiers or constructed identifying sets.
- **Entity Mapping**: combines the teleontology with corresponding data values in the datasets, generating the final KG.

6.1 Entity Matching

In our project, we gathered data from one single source, because it were sufficiently large to develop a KG respecting our purpose, this makes the entity matching problem trivial

6.2 Entity Identification

In this section we ensure thet each entity in the set is uniquely identified, we created specific identifiers. In some cases we already had data property used as IDs (like weather stations), in other (like the measurements), we created a specific ID, we want to highlight the entity of TimeInterval, which has an identifying set, composed by staring time and ending time.

6.3 Entity Mapping

Entity mapping integrates the teleontology with the datasets by defining the relationships between entity types and their data values.

We did so by Karma tool. The output of this phas is in Entity Definition of our repository, in the following images we can see a graphical representation

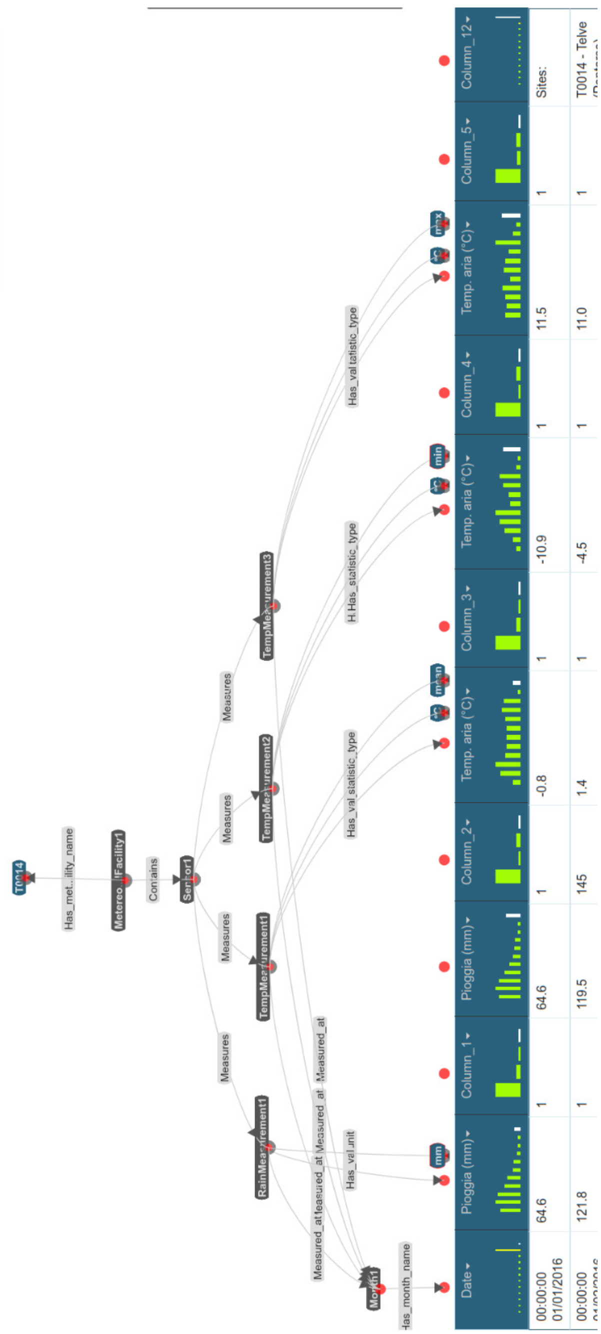
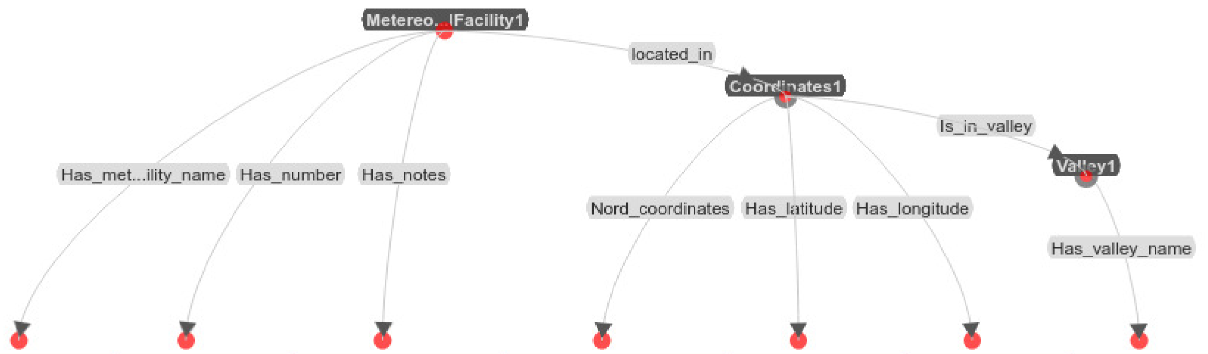


Figure 5: Visualization of the matching of the measurment to the teloeontology



Stazione	Tavoletta N	Note	coordinateEst/Nord	Latitudine	Longitudine	Bacino
T0101	32060010	ATTIVA - TP - POSIZIONI PRECEDENTI: dal 01/01/1935 al	660529/5113570	46°09'24.5" N	11°04'44.6" E	Bacino del fiume Adige
T0129	32 060100	ATTIVA - M - Palo vento 5 m	665157/5104242.5	46°04'18.5" N	11°08'08.5" E	Bacino del fiume Adige
T0135	32 060100	ATTIVA - M - Palo vento 10 m - Dal 30/04/2008 al 03/08/2012	662432/5106821	46°05'44.3" N	11°06'04.9" E	Bacino del fiume Adige
T0008	1 045050	MN - Gestita da IASMA - Palo vento 5 m - ID MTX: 002; ID DB_CM5800:	711553/5132107.7	46°18'34.3" N	11°44'51.0" E	Bacino del fiume Avisio
T0059	32 044160	ATTIVA - TP	700189.5/5126284.5	46°15'38.2" N	11°35'51.4" E	Bacino del fiume Avisio

Figure 6: Visualization of the matching of the weather station to the teloeontology

The output of this phase is a sequence of *ttl* files, one for each etype, containing all the data retrieved during the *Information Gathering* phase and the *Knowledge phase* that described the relevations made by the facilities and their geographical coordinates. These files can be fully reused in order to obtain informations through dedicated programs that will run SPARQL queries on them, as described in the next chapter

7 Evaluation

At the conclusion of the iTelos procedure we obtained a Knowledge Graph (KG) generated from external data of the real world; however it is important now to proceed by understanding if the result satisfies our needs and expected performances, for this it is necessary to make an evaluation of the KG, and in order to do it we have to rely on a software that interrogates our KG: such tool is represented by GraphDB, a program that allows to load the final KG and execute SPARQL queries on it in order to retrieve the necessary informations. In this final section we will cover three different aspects of the evaulation process:

- the general statistics of our KG
- the knowledge evaluation phase, in particular the different coverage types
- the data evaluation phase, foucsing mainly on the connectivity of the KG
- the conversion of the various Competency Questions in SPARQL queries, in order to check if our KG is able to satisfy them

7.1 KG statistics

As first point in the evaluation it is useful to report some statistics in order to have a better understanding of the situation:

- In total there are 17 etypes, and among them we count: *MeteorologicalMeasurement*, *TempMeasurement*,*Month*,*WindMeasurement*, *RainMeasurement*,*HumidityMeasurement*,*PressureMeasurement*, *SolarRadiationMeasurement*,*MeteorologicalFacility*,*Valley*, *Coordinates*,*Sensor*
- 12 data properties and 10 object properties

Both these statistics have been retrieved through following SPARQL queries that can be found here

7.2 Knowledge Layer evaluation

An important aspect of Knowledge Graphs is how well they are able to cover the portion of knowledge we want to incapsulate inside them, specifically this concept can be illustrated with the Competency Questions (CQs): if a particular Knowledge Graph has been designed well then it will be able to cover as many CQs as possible. We formalize this concept by saying that we want to compute the **coverage** of our teleontology T at Knowledge Level. Referencing the previous sections we can retrieve:

- the **Entity Types Coverage**, which measures the ratio of how many etypes are present in our teleontology:

$$Cov_E(CQ_E) = \frac{|CQ_E \cap T_E|}{|CQ_E|} = \frac{11}{18} = 61.1\%$$

  where T_e are the etypes in our generated teleontology, while CQ_e are the etypes related to the Competency Questions CQs.

- last, the **Property Coverage** it measures how many attributes in the Knowledge Layer are present in the Teleology. The formula is:

$$Cov_P(CQ_P) = \frac{|CQ_P \cap T_P|}{|CQ_P|} = \frac{12}{29} = 41.3\%$$

  where T_P are the properties in our generated teleontology, while CQ_P are the properties associated to the Comptetency Questions CQs.

It is also possible to define the coverage by analyzing the teleontology T generated in the previous sections, and comparing it with various knowledge elements called Knowledge Atoms (KATOMs) taken from the reference ontology generated in the knowledge phase:

- when the considered KATOM is an etype we talk about **Entity Type Coverage**, which measures the ratio of how many etypes are present in our teleontology:

$$Cov_e(KATOM_e) = \frac{|KATOM_e \cap T_e|}{|KATOM_e|} = \frac{12}{18} = 66.6\%$$

  where T_e are the etypes in our generated teleontology, while $KATOM_e$ are the etypes defined at knowledge level.

- similarly, we define the **Object Property Coverage** when the considered KATOM is about relationships, and it measures how many relationships in the Knowledge Layer are present in the Teleology. In this case the formula becomes:

$$Cov_{OP}(KATOM_{OP}) = \frac{|KATOM_{OP} \cap T_{OP}|}{|KATOM_{OP}|} = \frac{5}{12} = 41.6\%$$

  where T_{OP} are the relationships in our generated teleontology, while $KATOM_{OP}$ are the relationships defined at knowledge level.

- last, the **Data Property Coverage** treats KATOMs as attributes, and it measures how many attributes in the Knowledge Layer are present in the Teleology. The formula is:

$$Cov_{DP}(KATOM_{DP}) = \frac{|KATOM_{DP} \cap T_{DP}|}{|KATOM_{DP}|} = \frac{11}{13} = 84.61\%$$

  where T_{DP} are the relationships in our generated teleontology, while $KATOM_{DP}$ are the relationships defined at knowledge level.

Etypes	**MeteorologicalMeasurement**	**TempMeasurement**	**Month**	**WindMeasurement**	**RainMeasurement**	**HumidityMeasurement**	**PressureMeasurement**	**SolarRadiationMeasurement**	**MeteorologicalFacility**	**Valley**	**Coordinates**	**Sensor**
MeteorologicalMeasurement	36101	0	14541	0	0	0	0	0	0	0	0	0
TempMeasurement	0	17610	6106	0	0	0	0	0	0	0	0	0
Month	0	0	2806	0	0	0	0	0	0	0	0	0
WindMeasurement	0	0	2572	7240	0	0	0	0	0	0	0	0
RainMeasurement	0	0	2447	0	4776	0	0	0	0	0	0	0
HumidityMeasurement	0	0	1183	0	0	2247	0	0	0	0	0	0
PressureMeasurement	0	0	1141	0	0	0	2163	0	0	0	0	0
SolarRadiationMeasurement	0	0	1092	0	0	0	0	2065	0	0	0	0
MeterelogicalFacility	0	0	0	0	0	0	0	0	36101	0	24	23
Valley	0	0	0	0	0	0	0	0	0	24	0	0
Coordinates	0	0	0	0	0	0	0	0	0	24	72	0
Sensor	14541	6106	0	2572	2447	1183	1141	1092	0	0	0	0

All the SPARQL queries used to obtain the various Teleontology vs KATOM coverages can be found here. As we can see the Object Property Coverage results to be low, which means that there is not much correlation between the different etypes of the Knowledge Graph. One of the possible causes could be the usage of a single informal dataset for the building phase instead on relying on multiple sources. The Entity Type coverage also seems to have some gaps inside it, in fact we can identify the missing etypes through the following SPARQL query:

- ClimateChange
- Trentino
- TimeInterval
- ClimateChangeIndicator
- Trend

7.3 Data Layer evaluation

the knowledge layer evaluation gives us only a generic view on the situation: in fact it tells us how much knowledge is covered by the defined KG, however it does not provide any useful informations about the structure of the graph itself, like the connectivity between the nodes. The evaluation of the data layer aims to fill this gap by representing the Entity and Property connectivity through a **Connectivity Matrix**. In particular we define two metrics:

- the **Entity Connectivity** for a specific EType X, which represents how much entities are connected to each other: $EC(X) = \frac{\sum_{Y=1}^{N}(X,Y)}{OP(X)}$
- the **Property Connectivity** for a specific EType X, which tells us how much the entities are connected to their properties: $PC(X) = \frac{(X,X)}{DP(X)}$

Where (X, Y) is a cell in the connectivity matrix, $OP(X)$ is the number of object properties for the EType X, and $DP(X)$ is the number of data properties in the EType X. By calculating the number of object properties and data properties for each etype we can easily retrieve the Entity Connectivitie:

- *MeteorologicalMeasurement*: $\frac{36101+14541}{1} = 50642$
- *TempMeasurement*: $\frac{17610+6106}{1} = 23716$
- *Month*: 0
- *WindMeasurement*: $\frac{2572+7240}{1} = 9812$

- *RainMeasurement*: $\frac{4776+2447}{1} = 7223$
- *HumidityMeasurement*: $\frac{2247+1183}{1} = 3430$
- *PressureMeasurement*: $\frac{1141+2163}{1} = 3304$
- *SolarRadiationMeasurement*: $\frac{2065+1092}{1} = 3157$
- *MeteorologicalFacility*: $\frac{36101+24+23}{2} = 18074$
- *Valley*: 0
- *Coordinates*: $\frac{72+24}{1} = 96$
- *Sensor*: $\frac{14541+6106+2572+2447+1183+1141+1092}{1} = 29082$

Also we can find the Property Connectivity of each etype:

- *MeteorologicalMeasurement*: $\frac{36101}{4} = 9025$
- *TempMeasurement*: $\frac{17610}{4} = 4402$
- *Month*: $\frac{2806}{1} = 2806$
- *WindMeasurement*: $\frac{7240}{3} = 2413$
- *RainMeasurement*: $\frac{4776}{2} = 2388$
- *HumidityMeasurement*: $\frac{2247}{2} = 1123$
- *PressureMeasurement*: $\frac{2163}{2} = 1081$
- *SolarRadiationMeasurement*: $\frac{2065}{2} = 1032$
- *MeteorologicalFacility*: $\frac{36101}{4} = 9025$
- *Valley*: $\frac{24}{1} = 24$
- *Coordinates*: $\frac{72}{3} = 24$
- *Sensor*: 0

We can notice how the Property Connectivity of Sensor is set to zero since there is no instance of this etype in our Knowledge Graph. Both these metrices can be generalized to the whole knowledge graph:

- the **Entity Connectivity**: $EC(X) = \sum_{X=1}^{N} EC(X) = 148536$
- the **Property Connectivity**: $PC(X) = \sum_{X=1}^{N} PC(X) = 33343$

7.4 Queries

In previous sections we defined Competency Questions (CQs) that describes hypotetical scenario that our Knowledge Graph (KG) should be able to cover, which must be then turned into SPARQL queries and run on our teleontology in order to check it returns the predicted results. We can model the whole situation in the following way: for each CQ we describe the etypes and property values it should display, the query run on the KG and the actual results:

- The Competency Question Q1 query can be found here. From the theoretical point of view of our model the query should return the temperatures of different metereological stations in the period 2015-2025, and we can see that it actually returns data about this temporal window. Notice how the query considers only the mean, which means that some metereological stations are excluded
- the Competency Question Q2 query can be found here. The Competency Question asks for the distribution of the metereological facilities but since we cannot link directly the coordinates with a specific location we will just print the latitude and the longitude of the facility.
- The Competency Question Q3 query can be found here. The body of the request splits the data in two distinct periods, the one that goes from 2015 and terminates in the 2019, and the other one that goes from 2020 and terminates in 2025. After doing this it calculates the average temperature over these two periods separately and then it retrieves the temperature increase happened between the two eras. The expected result should display the increase temperature over different areas in Trentino region, however since no areas are provided we consider the metereological stations as standalone locations
- The Competency Question Q5 query can be found here. Also in this case we consider the metereological stations as distinct locations, which allows us to perform the query interrogation. The theoretical result should give us the different metereological stations, togheter with the statistics about the rainfalls, disposed in descendent order, and we can see that the actual query result actually satisfies the criteria
- The Competency Question Q6 query can be found here. The body of the request associates a number to each month in order to associate them with the season they belong, then it simply retrieves the interested data linked to a specific period. The expected result should return the months and the seasons over the considered decade with the highest solar radiation statistics, and we can see it actually satisfies the criteria
- The Competency Question Q8 query can be found here. The expected result should return the humidity trend, precisely the minimum,maximum and average values, over the summer of the last three years, which means that the query not only must consider only *monthNum* equal to six, seven and eight (respectively June,July and August) but also it must filter the years not included in the period 2023-2025, extremes included. As we can see the query actually satisfies the required criteria
- The Competency Question Q9 query can be found here. We ask to the Knowledge Graph the rainfalls trend of the last five years by simply filtering the time when the data have been

acquired and considering them only if they have relevations about the rain volumes, which means that the latter statistics in these cases must be greater than zero. The expected result should return the rainfall trend with the complete statistics, maximum, minimum and average rain volumes also in this case, and we can see how the query satisfies the required criteria

- The Competency Question Q10 query can be found here. The query checks if the station contains the humidity measurements we are looking for, then it filters the data of the period 2016-2025 (extremes included). Theoretically speaking it should return the humidity trends considering maximum, minimum and average values, togheter with the station that made the measurements and the year when the relevations have been produced, and we can see that the actual reult perfectly match the expectations

8 Metadata Definition

Throughout the development of the project there have been defined metadata, which allows the reusage of informations across other knowledge graphs. In this section we define four type of metadata:

- Language resources metadata
- Knowledge resources metadata
- KG metadata

8.1 Language resources metadata

Language metadata describe what kind of ontologies were used and classify them based on the role they have. The metadata include various fields:

- *Version*: the version of the ontologies used in the project
- *Prefix*: defines a shorthand string that replaces an Internationalized Resource Identifier (IRI)
- *Annotator*: the Annotator of the ontology
- *Owner*: tells us who currently has the property of the generated ontology
- *License*: the type of license applied for the current ontology
- *Other Namespaces Reused*: indicates if external namespaces have been reused
- *Generation DateTime*: indicates when the current ontology has been generated
- *Language*: this field contains the information about which natural language the ontology uses for the names of etypes and properties
- *Translators*: contains all the translations made to other languages

- *Keywords*: define the keywords used to summarize the role of the ontology
- *DatDomain*: contains the language domain on which the ontology holds
- *Validator*: tells us the identity of who should review the work of the Annotator
- *Reference (Tele)ontology*: defines the main teleontologies used.
- *Reference UKC Version*:
- *Project Page*: The domain type which this ontology belongs to
- *Type*: the type of metadata described

The complete metadata encoded as a spreadsheet can be found here

8.2 Knowledge resources metadata

Knowledge metadata describe the teleontology generated during the Knowledge phase, including some additional information that helps to frame the resource in a defined context. The fields of the metadata are the following:

- *Keyword*: a keyword that summarize the purpose of the teleontology
- *Publisher*: the publisher of the teleontology
- *Domain*: indicates the type of knowledge domain where the teleontology can be reused
- *Creator*: reports the complete names of the authors of the teleontology
- *PublicationTimestamp*: reports the timestamp that represents the exact moment when the teleontology has been created
- *Language*: describes the language on which the teleontology informations are encoded
- *Version*: the version of the teleontology
- *Owner*: reports the current owner of the teleontology. Notice that it does not necesseraly overlaps with the authors, because it is possible they give their teleontology to someone else
- *Level*: describe the level of abstraction that the teleontology represents
- *License*: indiecates the type of license applied to the teleontology
- *Type*: indicates the kind of type of resource we are describing

The complete metadata encoded as a spreadsheet can be found here

8.3 KG metadata

The final KG metadata describe informations about the project, its purpose togheter with various data about the authors in order to track who did what. The structure of the metadata will be the following:

- *License*: describe the license on which the knowledge graph is released
- *Category*: it is the real world context where the Knowledge Graph can be reused
- *Maintainer*: indicates the complete name of the current maintainer of the knowledge graph
- *Author(s)*: indicates the complete name of the creators of the knowledge graph
- *Author(s) Email*: reports the email of the authors of the knowledge graph
- *Tags*: describes various tags that describe the knowledge graph on KnowDive
- *Publication Date*: indicates the date when the knowledge graph has been released

The complete metadata encoded as a spreadsheet can be found here

9 Open Issues

As we saw during the whole procedure it is clear that the knowledge graph needs to be integrated with more data, in particular it is necessary to include all the metereological stations spread across the Trentino region and the locations of the different valleys and territories. As mentioned in the purpose formalization phase one of the major obstacles about the data retrievement has been the impossibility to scrape the meteotrentino dataset due to the fact that the download links were dynamically generated, making the manual retrievement the only possible solution; to fix this problem other datasets must be consulted in order to cover a wider range over the region, or at least find a way to automatize the scraping process from the resources illustrated in this report. Also the data about the geography of the single locations suffered of a similar problem, which requires a further extension of the Knowledge Graph through external resources.